ZHENGMIN PIAO

Hefei, China

Email: zmpiao@mail.ustc.edu.cn

RESEARCH INTERESTS

Statistical Inference, Stochastic Modeling, Optimization and Decision Methods, Machine Learning, and Data-Driven Systems, with applications to Industrial Analytics, Semiconductor Manufacturing Systems, Urban Systems, and Quantitative Finance.

EDUCATION

University of Science and Technology of China (USTC), Hefei, China Aug. 2023 – Present
B.S. in Statistics, Minor in Finance (Expected Graduation: Jun. 2027)

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant, Anhui Provincial Major Social Science Project Sep. 2025 – Dec. 2025
Supervisor: Prof. Jiaquan Yao

- Collected multi-year firm-level panel datasets (N=38) across China, US, Europe, and East Asia; applied econometric models to analyze R&D investment and innovation outcomes.
- Developed statistical models including fixed/random-effects panel regression, PCA, and ridge regression to handle high-dimensional and small-sample settings.
- Quantified the impact of patient capital on innovation performance using multivariate statistical analysis and financial feature engineering.

Intern, Organization Department, Yanbian Prefecture Government Jul. 2025 – Aug. 2025
Yanbian, Jilin, China

- Processed multi-year incident datasets (fires, traffic accidents, structural collapses, floods) using Python; developed optimization models for surveillance camera placement.
- Integrated GIS-based visibility analysis and spatial coverage modeling under terrain constraints; formulated a robust optimization framework considering camera reliability.
- Solved integer programming models using Gurobi/CPLEX; applied heuristic methods (Lagrangian relaxation, ADMM) for large-scale constrained optimization.
- Evaluated solutions using coverage, cost efficiency, and robustness metrics; delivered data-driven policy recommendations reducing system cost by approximately 40%.

SELECTED RESEARCH PROJECTS

Small-Sample Confidence Interval Comparison for Rare-Event Probabilities 2026
Independent Course Research Project

- Conducted a comparative study of statistical inference methods for rare-event probability estimation under small-sample and class-imbalance settings.

- Implemented classical and Bayesian interval estimation methods (Clopper–Pearson, Wilson, Wald, Bayesian credible intervals) in R.
- Evaluated methods via Monte Carlo simulations in terms of coverage probability, interval width, and finite-sample robustness.

Counterfactual Analysis of Voting Rules Using Constrained Bayesian Inference 2026
Mathematical Contest in Modeling (MCM/ICM), Problem C

- Developed a Bayesian inference framework with truncated Dirichlet priors to estimate latent voting distributions under constrained settings.
- Designed counterfactual simulation experiments to evaluate voting rule sensitivity and uncertainty propagation.
- Built mixed-effects and survival models to analyze heterogeneous decision behavior across judges and fans.

HONORS & COMPETITIONS

Meritorious Winner, Mathematical Contest in Modeling (MCM/ICM), Problem C 2026
First Prize, Anhui Division, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2025
Third Prize, Anhui Division, National Undergraduate Mathematics Competition 2024

VISITING & EXCHANGE PROGRAMS

Global Leadership Visit and Exchange Program, Singapore Management University 2025
Project: AI-driven quantitative investment strategy design using convolutional neural networks and NLP-based sentiment analysis on financial data.

LEADERSHIP

President, Student Union, School of Management Apr. 2025 – Apr. 2026
Student Representative, AI Computing Platform (USTC) Apr. 2026 – Present

SKILLS

Programming & Data Science: Python, R, SQL, Julia, C++
Optimization Tools: Gurobi, CPLEX, Stata
Languages: Chinese (native), Korean (native-level), English (fluent)